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RESEARCH-ARTICLE

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Phishing Website Detection on Imbalanced Datasets Based on Diffusion Model and Multi-Head Self-Attention

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Abstract

With the rapid development of the Internet, phishing websites, as a form of malicious network activity, have become one of the serious threats in the field of network security. Phishing websites will disguise themselves as formal and legal websites and lure users to provide sensitive personal information, resulting in significant economic and privacy losses. In order to effectively deal with this threat, our research aims to improve the efficiency of phishing website detection, with a special focus on describing the impact of unbalanced data samples on the effectiveness of traditional detection methods. To address this challenge, we have introduced a new approach called the “Probability Classification Generator”(PCG). Drawing inspiration from generating adversarial networks, PCG solves the problem of data imbalance. The empirical research results highlight significant improvements in accuracy and recall indicators, highlight significant improvements in alleviating data imbalance issues and significant enhancement in the detection capabilities of phishing websites.

CCS Concepts

• **Security and privacy** → Network security; Denial-of-service attacks.

Keywords

Diffusion Model, Attention Mechanism, Convolutional Neural Network, Data Sample Imbalance, Phishing Website Detection

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1 Introduction

Phishing is an online social engineering technique in which attackers pretend to be well-known institutions to trick users into revealing sensitive data such as credit card numbers and passwords. Such attacks can lead to serious consequences, including identity

theft, privacy breaches and financial loss. The Anti-Phishing Working Group (APWG) reported a record 222,127 phishing incidents in June 2021 [1], and nearly 5 million incidents were recorded in 2023—the highest annual total to date [2]. The IBM Security X-Force team’s Threat Intelligence Index (2022) also highlights a surge in network attacks during this period [3]. The IBM Security X-Force team’s Threat Intelligence Index (2022) also highlighted the surge in cyberattacks during this period [4].

Responses to phishing include things like Google’s blacklist, which collects reported phishing sites and integrates them directly into Chrome [5]. Similarly, vendors like McAfee and Panda Security have added dedicated anti-phishing modules to their antivirus suites [6, 7]. Traditional methods that rely on blacklists and whitelists are still ineffective in detecting zero-day phishing attacks [8, 9].

To address these shortcomings, researchers have begun to use machine learning methods such as naive Bayes, random forests, and support vector machines to extract features from uniform resource locators and page content. [10]. While these methods are effective to some extent, they rely on hand-crafted feature sets, which can introduce bias. Deep learning technologies, including Long-Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNN), and Deep Belief Networks (DBN), can automatically perform feature extraction, but often fail when faced with highly unbalanced datasets, where the number of legal URLs far exceeds the number of phishing URLs. [11–14].

This paper introduces an innovative phishing detection approach that combines diffusion models with multi-head self-attention mechanisms. Key contributions include:

1. A diffusion-based data generation mechanism to address class imbalance by creating high-quality synthetic samples;
2. A feature extraction framework combining multi-head self-attention and CNNs to capture phishing-specific features;
3. The Probabilistic Classification Generator (PCG), which integrates diffusion models with adversarial learning, achieving superior precision, recall, and stability; and
4. Architectural and training optimizations for real-world applicability.

Experimental results demonstrate that this approach consistently outperforms existing methods on multiple benchmark datasets, validating its effectiveness and applicability.

2 Related Works

Traditional phishing detection techniques rely on hand-crafted metrics such as URL patterns, domain name characteristics, and page content attributes, and use rules-based engines or threshold checks



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to flag suspicious websites. Sengar P and others launched Phish-Guard, a browser toolbar for checking URL structure and content characteristics [15]. Rao and Pais conducted research to evaluate which of these features are most effective for detection, while Mohammed and others created an automated framework to evaluate a large number of features to distinguish phishing websites from legitimate websites [16]. Although these methods are straightforward to deploy, their static nature makes them easy to circumvent by attackers who constantly adjust their strategies.

To improve the accuracy and reliability of detection, the researchers used machine learning methods that can automatically identify distinguishing patterns between phishing websites and legitimate websites. Dol and others applied association classification techniques to mine and identify significant feature differences [17], and Gupta studied feature subset selection and combined multiple learning algorithms to improve performance [18]. Aamir emphasized the key role of feature engineering by systematically evaluating the impact of feature engineering on the effectiveness of classifiers [19]. Although these machine learning models perform well when obtaining sufficient data and carefully designing features, their success is still closely linked to the quality of feature extraction, and poor feature design can significantly reduce detection rates.

Deep learning technology further advances phishing detection by eliminating the burden of a large amount of manual feature engineering. By automatically learning hierarchical representations, these models are able to discover subtle patterns that traditional methods miss. Vinayakumar demonstrated that deep models more effectively capture implicit URL patterns than classical approaches [20]. Recently, Alsaedi M combined graph neural networks with multimodal learning to integrate URL structure, page content, and visual information to improve detection accuracy, and in another study they introduced a multi-head self-attention model that focuses on the most significant URL and page features, significantly improving results [21]. Despite these results, deep learning solutions often result in higher model complexity, longer training times, and huge computing requirements.

Recent work on phishing detection has introduced Generative Adversarial Networks (GANs) to enhance robustness and accuracy. Douzas G used conditional generation versus networks to explore how data imbalances affect vulnerability situations, and created various synthetic phishing examples to demonstrate the value of generation versus network-driven data enhancement [22]. Kumar V extended this approach by developing a semi-supervised conditional generation adversarial network based on game theory principles, optimizing its loss function to improve the discrimination between benign and malicious URLs [23]. Li Y and Zou W. combined Generative Adversarial Networks (GANs) with Convolutional Neural Networks (CNN) and Long Short-Term Memory Networks (LSTM) layers to train more powerful detection models using antiversarial generated phishing samples [24]. Together, these studies show that even though challenges in training stability and architectural complexity of Generative Adversarial Networks (GANs) remain to be resolved, Generative Adversarial Networks (GANs) have great potential to model complex phishing behavior through high-quality synthetic data and adversarial training.

Diffusion models have recently emerged as an effective generation tool to mitigate data imbalances [25]. By creating additional samples, they can increase the diversity of data sets and better capture complex data distributions. In our work, we apply this capability to phishing detection through the Probable Classification Generator (PCG), which combines a diffusion model with an attention mechanism. This fusion not only makes up for the category imbalance, but also expands and refines the feature space, allowing models to focus on the most information-valuable attributes. PCG provides higher accuracy and recall rates than traditional methods, providing a more reliable solution for identifying phishing threats.

3 Methodology

3.1 Generator

We first define a character set that contains all common URL components, including upper and lower case letters, numbers, and typical symbols, and then assign each character a unique integer index. Each URL is converted to lower-case form, then checked to ensure that each character belongs to the character set we define, and then evaluated against predefined length limits. Valid URLs are converted to a sequence of integer indexes and then padded with zeros or truncated as needed to match the fixed length. Finally, this one-dimensional index sequence is reshaped into a two-dimensional matrix for subsequent processing.

To create a grayscale image, we normalized each matrix element to a range of 0 to 255 by dividing its integer index by the maximum index (66) and then multiplying it by 255. This converts the character-based index into pixel intensity, allowing the digital representation of the URL to be rendered into a standardized grayscale image. Since each URL is converted to a fixed-size matrix, the generated images have the same size, providing a unified input to the diffusion model. In this way, each URL is represented both digitally and visually, ready for model training and analysis.

Diffusion models draw on the concept of diffusion processes in physics and apply it to data generation tasks in machine learning. Its basic philosophy revolves around creating high-quality data distribution through two interconnected processes: the forward process and the reverse process.

Forward Process Model: This model outlines a process that incrementally introduces noise to real data x_0 :

$$q(x_t | x_0) = N(x_t; \sqrt{\bar{a}_t} x_0 + (1 - \bar{a}_t) I) \quad (1)$$

In this context $\bar{a}_t$ constant hyperparameter, x_t represents the data after noise has been added, and I stands for the identity matrix.

Sampling Formula: By applying the reparameterization trick, we can draw samples from the forward process model:

$$x_t = \sqrt{\bar{a}_t} x_0 + \sqrt{1 - \bar{a}_t} \epsilon_t, \epsilon_t \sim N(0, I) \quad (2)$$

In the formula, ϵ_t represents the final noise distribution. It conforms to the Gaussian distribution

Reverse process model: This model is used to reverse the forward process. After the reverse process, data similar to the original data will be generated.

$$P_0(x_{t-1} | x_t) = N(\mu_0(x_t); \sum_0(x_t)) \quad (3)$$

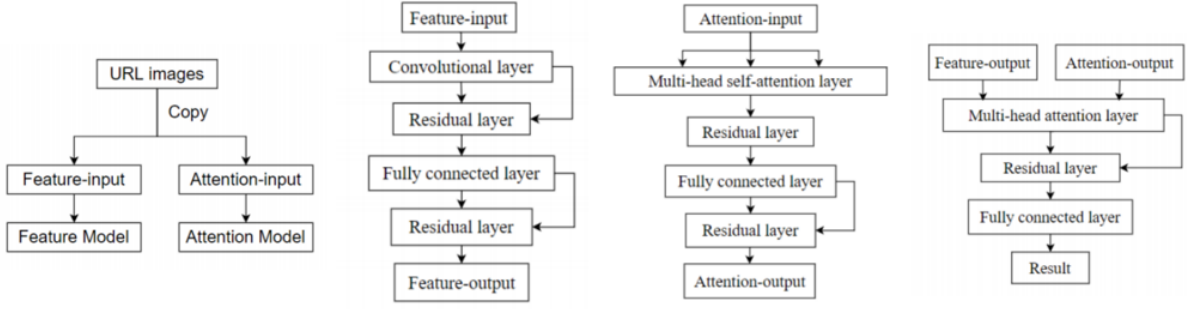


Figure 1: Modules of the MH-CNN model

To facilitate denoising during the reverse process, efficient network structures like UNet or ViT are often used, which perform an excellent job in extracting feature information from data. By using DDPM (Diffusion Driven Probability Model), we can infer the probability distribution at each step of the noise removal process and gradually restore the actual content of the data.

When the noise during the diffusion process is minimum and has a Gaussian distribution, the transition in the sampling chain can be modeled as a conditional Gaussian distribution. This enables this to be achieved using direct neural network parameterization.

3.2 Discriminator

Inspired by human visual perception, the attention mechanism solves data processing problems by selectively focusing on some information and ignoring others. This concept allows models to dynamically prioritize key information during processing and analyze less important information later. Input can be represented in the form of key-value pairs. In this setting, the “key” is used to calculate the attention distribution, while the “value” is responsible for generating the selected information content. Here is the formula for calculating the multi-head attention model.

MH-CNN introduces a multi-head attention mechanism based on Convolutional Neural Networks (CNN), allowing it to better process sequence data and capture global context information. In experiments, we found that MH-CNN has obvious advantages in processing sequential data, especially for URL or text data in the field of cybersecurity, because it can better capture key features and patterns, thereby improving the performance of the discriminator.

MH-CNN model in the paper comprises four primary modules: Input Module, Feature Module, Attention Module, and Output Module. As shown in Figure 1.

In the input module, the grayscale image of the URL is copied into two, one of which is forwarded to the features module, and the other is directed to the attention module. The feature module integrates convolution layers, residual layers and fully connected layers to extract feature data from URL features, uses multiple convolution kernels to explore different feature spaces, uses maximum pooling to capture the most important features, and adds residual layers to alleviate network depth. Performance degradation may be caused by increase, and the final feature output is linearly transformed through the fully connected layer. The attention module

uses a self-attention layer to calculate feature weights and learns the internal dependencies of input data to achieve comprehensive feature representation. Similar to the feature module, the attention module incorporates a residual layer and a fully connected layer to mitigate performance degradation and perform linear transformation. In the output module, the outputs of the attention module and the feature module are merged together. This module contains a self-attention layer to calculate the final feature weights, a residual layer and a fully connected layer for linear transformation. Classification judgment is done through a Sigmoid function, which gives the final classification decision.

By integrating these modules, the classifier effectively captures and processes the intricate features of URLs, enhancing the accuracy of phishing website detection.

3.3 Probabilistic Classifying Generator model

Inspired by the generative adversarial model (GAN), we tried to use the diffusion model as the generator and multi-head self-attention CNN as the discriminator to build an adversarial generative network, and referred to as the Probabilistic Classifying Generator (PCG). Figure 2 shows the Flowchart of the PCG model.

The training process of the PCG model first randomly initializes the parameters of the generator and discriminator. Then the generator uses a random noise vector to generate similar data samples. Then the discriminator receives the generated samples and the real samples for classification. The losses of the generator and the discriminator are calculated based on the classification results of the discriminator, and then the parameters are updated using the back propagation algorithm. The loss functions of the generator and the discriminator are as follows:

$$Loss_G = -E [\log D(G(z))] \quad (4)$$

$$Loss_D = -E [\log D(x) - E [\log (1 - D(G(z)))]] \quad (5)$$

After that, training will continue to iterate until stopping conditions are met, such as loss of convergence or reaching a predetermined number of training rounds. Finally, the model is trained, as are the parameters of the generator and discriminator, which can be used to generate realistic network data samples or perform threat detection tasks.

By combining the generative discriminant framework of the diffusion model with a multi-head self-attention convolutional neural

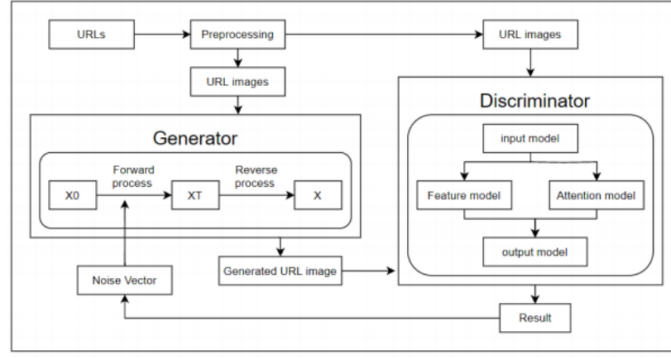


Figure 2: The Rec of different A or F.

Table 1: The result of PCG on different datasets.

	Kaggle	PhishTank	AWPG
Accuracy	95.46%	92.46%	93.81%
Recall	93.59%	88.95%	91.77%
Precision	94.57%	92.88%	93.39%
F1-score	93.26%	87.69%	91.84%

network, the PCG architecture makes full use of adversarial dynamics to improve detection accuracy and reliability. Diffusion components are capable of synthesizing diverse and high-fidelity samples, while self-attention mechanisms separate and highlight the most informative features. Together, these elements improve the robustness and practical performance of the PCG model in real-world phishing detection tasks.

4 Experiments

4.1 Datasets

We obtain data from publicly available standardized Kaggle datasets that collectively contain more than 500,000 URLs from both benign and malicious sources. We randomly selected 50,000 URLs with no more than 200 characters in length from this data pool. These URLs were then divided into a training set of 40,000 URLs and a test set of 10,000 URLs. This partitioning provides sufficient examples for training the PCG model, while retaining an independent test set to objectively evaluate its detection performance.

4.2 Evaluation Metrics and Analysis

In order to verify whether the generator (diffusion model) showed significant performance differences on different datasets, we selected Kaggle and PhishTank from different sources as two datasets, and compared their impact on generator performance through ablation experiments. As shown in Table 1. After training and testing the generator model, we introduced indicators such as accuracy (Acc), false alarm rate (FPR), and recall rate (Rec).

On the Kaggle dataset, the generator achieved very high accuracy, recall, accuracy, and F1 scores. This is likely due to the diversity and broad coverage of the dataset, allowing the model to learn a

wide variety of URL patterns and characteristics. In contrast, the performance of all indicators on the PhishTank dataset was slightly lower. Since PhishTank only contains confirmed phishing URLs, the malicious indicators of these URLs may be more subtle and diverse, making it more challenging for the generator to summarize. Compared to Kaggle’s results, both accuracy and F1 scores dropped, reflecting the more deceptive and covert nature of fishing samples in the real world.

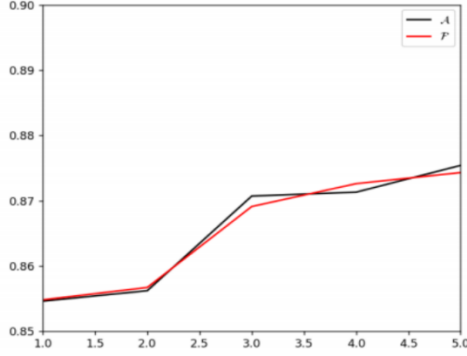
We studied how changing the number of attention modules and feature modules affects the performance of MH-CNN. In MH-CNN, the attention module is used to capture key information and learn global context, while the feature module is used to extract local features and patterns. By changing the number of attention modules and feature modules, we examined their impact on the performance of the discriminator. Experimental results show that increasing the number of attention modules can enhance the ability of MH-CNN to capture global information, thereby improving the performance of the discriminator. Increasing the number of feature modules can enhance the ability of MH-CNN to capture local features, thereby improving the accuracy and robustness of the discriminator.

In this experiment, we used the Kaggle dataset for training and conducted experiments with different numbers of attention blocks (A) and feature blocks (F) to observe their impact on performance. Table 2 shows the experimental results for different A and F values. The results showed that as the A and F values increased, both the accuracy rate (Acc) and the recall rate (Rec) increased. When both A and F are set to 5, the performance of the classifier is optimal.

Adding A and F’s can also lead to significant training costs. Data shows that when A and F exceed 3, both accuracy and recall remain almost constant. This shows that moderate increases in A and F can significantly improve classifier performance without excessively

Table 2: The classification result of different A and F.

A Blocks	F Blocks	Accuracy	Recall
1	1	89.65%	84.18%
2	2	90.24%	86.73%
3	3	91.87%	97.21%
4	4	93.46%	88.04%
5	5	93.51%	88.11%

**Figure 3: The Rec of different A or F.**

consuming resources. We chose to set the values of A and F to 3 in other experiments. We also conducted experiments with different values of A or F on the Kaggle platform. Figure 3 shows the Rec corresponding to different values of A when F is fixed to 3, and the Rec corresponding to different values of F when A is fixed to 3. The results show that Rec gradually improves with the increase of A or F. This phenomenon suggests that the performance of the model will improve with the increase of network depth. This improvement may be due to the greater number of parameters in the deeper network, which enhances the model’s fitting ability.

This article uses detailed experimental data comparison and analysis to deeply explore the specific performance of PCG model and GAN model in the field of network threat detection. Through systematic data verification and comparative evaluation, we can clearly understand the performance differences between the two models in key performance indicators such as detection accuracy, precision, and recall rate. The following is our detailed analysis and conclusions. Table 3 shows the experimental results.

In this study, we conducted a comparative analysis of the PCG model and traditional models to evaluate its performance in phishing website detection. The results show that PCG outperforms all other models in terms of key performance indicators.

PCG has shown significant advantages in accuracy, and its performance surpasses traditional models such as GAN and CNN, which highlights its excellent classification ability when detecting phishing websites. In terms of recall rate, PCG has performed well, even if there are category imbalances. The recall rate can still be maintained high and phishing URLs can be effectively identified. Compared with other models, PCG is significantly better at detecting these key positive instances. PCG surpasses other models in terms of accuracy, which shows that it can effectively reduce false positives and ensure reliable detection. This improvement highlights PCG’s ability to accurately distinguish phishing websites from legitimate websites, thereby enhancing the reliability of detection. Finally, PCG performed well in F1 scores, reflecting its ability to balance accuracy and recall rates. The improved F1 score shows that PCG can maintain both high accuracy and strong recall, thereby improving overall detection performance.

Compared to graph convolutional networks (GCN), the PCG model traded a slight reduction in recall in exchange for significant improvements in accuracy, precision, and F1 scores, resulting in better overall performance. The advantage of PCG is its ability to adapt to unbalanced data, which improves its effectiveness on all evaluation indicators. Compared with generative adversarial networks (GAN)-based methods, PCG also shows significant advantages. By combining adversarial learning, diffusion-based synthesis, and multi-head self-attention mechanisms, it can achieve higher detection rates with lower computing overhead. Overall, our experiments confirm that PCG is significantly superior to traditional deep learning and classic machine learning models in phishing detection, and performs well in terms of accuracy, recall, and F1 scores, especially in dealing with category imbalance issues.

Table 3: Comparison results between PCG and GAN.

	PCG	GAN	SVM	CNN	LSTM	GCN
Accuracy	96.14%	92.26%	90.15%	93.50%	94.10%	94.71%
Recall	95.50%	85.87%	88.09%	91.47%	92.83%	95.22%
Precision	94.84%	90.47%	86.50%	92.10%	93.20%	93.81%
F1-score	93.31%	89.46%	89.27%	90.75%	92.03%	92.50%

5 Conclusion

This study used a diffusion model to generate synthetic phishing URLs, creating a balanced training set that ensures adequate representation of malicious samples. Based on this dataset, I developed a probability classification generator (PCG) that combines diffusion-based synthesis, convolutional neural networks, and self-attention mechanisms for phishing detection. My experiments have shown that training on a diffusion-enhanced balanced dataset results in better detection performance than using only real URLs. When evaluated on the same test set, the probability classification generator (PCG) outperformed all other comparison models. The results of this study open up a new direction in the field of phishing website detection. We plan to further expand our work in the following areas. We will consider introducing more diverse feature extraction methods to enhance the adaptability of the model to new phishing URLs. We hope to explore the possibility of multimodal data fusion to improve the accuracy and robustness of detection by combining text, images and metadata.

I used tools to touch up the whole text. After using these tools, I reviewed and edited the content as needed and take full responsibility for the content of the publication.

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